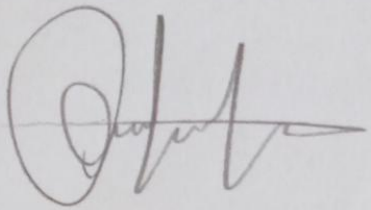
On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issue

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Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ some of the elimination

matrices are given below :

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a- Find the unknown entries $\alpha, \gamma, \beta, k, m = ?$

b- Find the eliminated upper triangle matrix $U = ?$

c- Determine the remaining elimination matrice.

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

$$\begin{vmatrix} (-1-\lambda) & 1 \\ 1 & (-1-\lambda) \end{vmatrix} = 0$$

$$\det(A - \lambda I) = 0$$

$$(-1-\lambda)^2 - 1 = 0$$

$$(-1-\lambda)^2 = 1$$

$-1-\lambda = 1$ $\quad$ $-1-\lambda = -1$

$\boxed{\lambda_2 = -2}$ $\quad$ $\boxed{\lambda_1 = 0}$

$\lambda_1 = 0$ için

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda_2 = -2$ için

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot X_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = c_1 \cdot \lambda_1 X_1 + c_2 \cdot \lambda_2 \cdot X_2$$

$$= 0 + c_2 \cdot (-2)^n \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = 0$$

Bu denklem doğru

Son soru

$$\begin{bmatrix} 1 & 8/3 & -4 & 8/3 \\ 0 & -49/3 & 8y-9 & 64/3 \\ 0 & 49/3 & 3+2y & 16/3 \\ 0 & 343/3 & y-6 & -2/3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 8/3 & -4 & 8/3 \\ 0 & -49/3 & 8y-9 & 64/3 \\ 0 & 0 & 12-6y & \frac{208}{9} \\ 0 & 0 & \frac{321y-600}{49} & \frac{250}{441} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 0 \end{bmatrix}$$

3. satırı $-1/2$ ile çarpıp 4. satır ile toplamak.

$$-\frac{1}{2}(12-6y) = \frac{321y-600}{49} \qquad y = \frac{51}{29}$$

$$\begin{bmatrix} 1 & 8/3 & \frac{-51}{29} & 8/3 \\ 0 & \frac{-49}{3} & \frac{147}{29} & \frac{64}{3} \\ 0 & 0 & \frac{42}{29} & \frac{208}{9} \\ 0 & 0 & 0 & \frac{-4846}{441} \end{bmatrix}$$

$R_2 \to \frac{-3}{49}$

$R_3 \to \frac{29}{42}$

$R_4 \to -\frac{441}{4846}$

Grp

$$\begin{bmatrix} 1 & 8/3 & -51/29 & 8/3 \\ 0 & 1 & -9/29 & \frac{64}{49} \\ 0 & 0 & 1 & \frac{5016}{1421} \\ 0 & 0 & 0 & 1 \end{bmatrix} = U$$

Note

$$U = \begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

upper triangular matris

$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ olması gerekir bu

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$

R_1 yi -8 ile çarpmışız. R_2 ile toplamışız.
3. satırı m ile çarpmışız yok etmek için
4. satırı (-1) ile çarp.

m ile çarp $\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9 & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$m = -2$

$3\alpha = 8$

$\alpha = 8/3$

Son durum

$\begin{bmatrix} 1 & 8/3 & -4 & 8/3 \\ 8 & 5 & -9 & 1 \\ 2 & \beta & 3 & 64/9 \\ 1 & 14 & -6 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 8/3 & -4 & 8/3 \\ 0 & -49/3 & 8y-9 & -64/3 \\ 0 & \beta - 16/3 & 3+2y & 16/3 \\ 0 & 34/3 & y-6 & -2/3 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$

2. satırı (-1) ile çarp 3. satır ile topla
2 satırı k ile çarpıp 4. satırla topladık.

$\frac{49}{3} = \beta - \frac{16}{3}$

$\beta = \frac{65}{3}$

$k \cdot \left(\frac{-49}{3}\right) = \frac{34}{3}$

$k = \frac{34}{49}$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

$$\det(A - \lambda I)$$

$$\begin{bmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{bmatrix} = (2-\lambda)(1-\lambda) = 2$$

$$2 - 3\lambda + \lambda^2 = 2$$

$$\lambda^2 - 3\lambda = 0$$

$$\lambda(\lambda - 3) = 0$$

$$\lambda = 0 \quad \lambda = 3$$

$\lambda = 0$ için

$$A - \lambda I = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \cdot x_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = 3$ için

$$A - \lambda I = \begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix} x_4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_4 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Altta yaptığım işlemde C_2 sayısı negatif çıkıyor. Ama formülde

$$C_1 \cdot \lambda_1 \cdot x_1 + C_2 \cdot \lambda_2 \cdot x_2$$

$$C_1, C_2 \in +\mathbb{R}$$

$$\boxed{C_1, C_2 > 0}$$

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

$$\begin{vmatrix} (\lambda+2) & 1 \\ 1 & (\lambda+2) \end{vmatrix}$$

$$|\lambda I - A| = 0 \qquad (\lambda+2)^2 = 1$$

$$(\lambda+2)^2 = 1$$

$\lambda = -1$ için

$\lambda = -1$ c_1 ile c_2 katı katı

$$\lambda I - A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x_s = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_s = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = c_1 \cdot \lambda_1 \cdot x_s + c_2 \lambda_2 \cdot x_s$$

$$c_1 (-1)^n \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \cdot (-1)^n \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$c_1 + c_2 = 1$$

for $n = 0$ $\begin{bmatrix} v_k \\ w_k \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$c_1 - c_2 = 1$$

$$c_1 + c_2 = 0$$

$$3c_2 = -1$$

$c_2 = -\frac{1}{3}$ $\quad c_1 = \frac{1}{3}$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = 0{,}33 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0{,}33 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$k \to \infty$ $\quad (-1)^n \to \infty$

$$\begin{bmatrix} v_\infty \\ w_\infty \end{bmatrix} = \begin{bmatrix} 0{,}33 \\ -0{,}33 \end{bmatrix}$$